

Safety Data Sheet

Issued Date: September 18, 2008

Revised Date: September 25, 2008

5. FIRE-FIGHTING MEASURES

EXTINGUISHING MEDIA Dry powder, alcohol-resistant foam and carbon dioxide extinguisher, dry sand, water spray

SPECIFIC HAZARDS ARISING FROM THE CHEMICAL May produce poisonous and irritated gasses upon a fire.

SPECIFIC FIRE FIGHTING MEASURES Workers should wear appropriate protectors (glasses, cloths, mask for poisonous gasses, etc.), then extinguish should be performed up wind.

6. ACCIDENTAL RELEASE

PERSONAL PRECAUTIONS, PROTECTIVE EQUIPMENT AND EMERGENCY PROCEDURES

Wear appropriate protective equipment (refer to 8. Exposure Control/Personal Protection) to avoid contact to eyes, skin and inhalation.

ENVIRONMENTAL PRECAUTIONS, RECOVERY/NEUTRALIZATION

Take precaution to prevent product to flow into rivers and affect environment.

METHODS AND MATERIALS FOR CONTAINMENT AND CLEANING UP

In case of a small spill, absorb with dry sand, soil, sawdust, cloth, etc., then place in a sealable container.

In case of large spills, dike and prevent overflow. Guide to a safe place then dispose properly.

SECONDARY ACCIDENT PREVENTION MEASURE

All ignition sources should be quickly removed. (No smoking in vicinity, prohibit sparks or fire sources)

7. HANDLING AND STORAGE

HANDLING

ENGINEERING MEASURES

Wear protective equipment. Perform engineering measures in accordance with 「8. Exposure Control / Personal Protection」.

**LOCAL VENTILATION
/GENERAL VENTILATION
SAFETY HANDLING
PRECAUTIONS**

Perform local and general ventilation in accordance with 「8. Exposure Control/Personal Protection」.

Fire strictly prohibited

STORAGE

ENGINEERING MEASURES

Keep container tightly closed. Protect from direct sunlight. Store the product at moderate temperature.

Refer to the technical data, specifications, and a product label about handling range of temperature.

**CONTAINER AND PACKAGING
MATERIALS**

Keep only in original container. Do not transfer the product to another container.

8. EXPOSURE CONTROL / PERSONAL PROTECTION

CONTROL PARAMETERS

Silica

ACGIH TLV

OSHA PEL

Toluene

Not established
50 ppm

Not established
100 ppm

ENGINEERING MEASURES

If handling this product indoors, seal off sources or use a local mechanical ventilation system, etc.

Place a safety shower, hand washing sink and an eye bath near work area and clearly marked.

PERSONAL PROTECTION EQUIPMENT

RESPIRATORY PROTECTION

Wear mask to prevent organic gas poisoning, if necessary.

HAND PROTECTION

Wear appropriate protective gloves (Polyethylene, rubber, etc., solvent impervious materials).

EYE PROTECTION

Use eye protection. (preferably goggles)

SKIN AND BODY PROTECTION

Wear personal protection apron, boots, if necessary. Do not work with short sleeve shirts.

SANITARY MEASURES

Wash hands thoroughly after handling. Do not eat, drink or smoke when using this product.

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9. PHYSICAL AND CHEMICAL PROPERTIES**PHYSICAL STATE**

APPEARANCE	Liquid
COLOR	Red
ODOR	Distinctive odor
FLASHPOINT	Non-flammable
AUTOIGNITION TEMPERATURE	Over 200°C
EXPLOSION LIMIT	No data
SPECIFIC GRAVITY (DENSITY)	1.11
SOLUBILITY	Slightly soluble in water
VISCOSITY	650 mPa·s

PHYSICAL STATE as Silica

MELTING POINT/FREEZING POINT	1710 °C, 1600-1750 °C(Sublimation at 1750 °C)
BOILING POINT	2230°C
(INITIAL AND RANGE)	
SPECIFIC GRAVITY (DENSITY)	2.65(20°C)
SOLUBILITY	Water: 0.2g(100ml, 3N Ammonia water, 18 °C) Slightly soluble in water

PHYSICAL STATE as Toluene

MELTING POINT/FREEZING POINT	-95°C
BOILING POINT	111°C
(INITIAL AND RANGE)	
FLASHPOINT	4.4 °C(Closed), 7.2 °C(Open)
SPECIFIC GRAVITY (DENSITY)	0.87(20 °C)
SOLUBILITY	Insoluble in water, Soluble in Ethanol, Ether, Acetone, Benzene

10. STABILITY AND REACTIVITY

STABILITY	Stable under normal handling.
POSSIBLY HAZARDOUS REACTION	Reacts upon exposure to heat, metal and absence of oxygen.
CONDITION TO AVOID	Heating.
INCOMPATIBLE MATERIALS	Metal.
HAZARDOUS DECOMPOSITION	Incineration may produce poisonous gasses (Carbon monoxide, NOx, SOx, Low molecule organic compounds, etc.) upon condition.

11. TOXICOLOGICAL INFORMATION**HEALTH HAZARDS**

ACUTE TOXICITY	No data as product
SKIN CORROSION/IRRITATION	No data as product

Information on GHS Hazard Communication is in accordance with Japanese Law

12. ECOLOGICAL INFORMATION**ENVIRONMENTAL HAZARDS**

HAZARDS TO THE AQUATIC ENVIRONMENT	No data as product
ECOTOXICITY	No data

Information on GHS Hazard Communication is in accordance with Japanese Law

13. DISPOSAL CONSIDERATIONS

METHOD OF DISPOSAL	To dispose product, solicit waste disposal management experts. Prohibited that to dispose the waste or waste liquid containing the product in the river, etc., to reclaim, to dump the product as it is.
CONTAMINATED CONTAINERS AND PACKAGING	Handle in used container and cloth same as above.

14. TRANSPORT INFORMATION**INTERNATIONAL REGULATION**

SEA TRANSPORTATION	Not applicable
UN number	Not applicable
AIR TRANSPORTATION	Not applicable
UN number	Not applicable

JAPAN DOMESTIC REGULATION

LAND TRANSPORTATION	In accordance with Japanese Law
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15. REGULATORY INFORMATION

Handle in accordance with applicable laws and regulations.

16. OTHER INFORMATION

Portions of the above evaluation of dangerous and harmful effects may be insufficient, please perform adequate investigation.

The content in this report is based on information which was available as of the Effective date.

But Three Bond Co., Ltd. and its affiliates are not responsible for guaranteeing the above data and evaluations.

The above data assumes usage under normal working conditions.

In case of special handling is required, please handle with suitable safety measures according to the application and usage.

The content in this report may change due to new evaluation and tests, etc.

In case there are differences in the translation, the Japanese language version takes precedence.